Dataset Academic Summary 02 — PhonemeDF

PhonemeDF: A Synthetic Speech Dataset for Audio Deepfake Detection and Naturalness Evaluation

Nallaguntla et al., LREC 2026

The paper *PhonemeDF: A Synthetic Speech Dataset for Audio Deepfake Detection and Naturalness Evaluation* by Nallaguntla and colleagues (2026) presents a new dataset for researching the acoustic differences between real speech and synthetic speech at the phoneme level. The study focuses on a limitation of many approaches to voice forgery detection, which are based on classifying whole recordings without necessarily examining in which phonetic units the signs of forgery appear. The researchers seek to examine whether synthesis systems struggle to produce certain phonemes, and whether these differences can be exploited for detecting fake speech.

For this purpose the PhonemeDF dataset was built, based on 28,539 real recordings from LibriSpeech, alongside 199,773 synthetic recordings created by means of four Text-to-Speech (TTS) systems and three Voice Conversion (VC) systems. The dataset includes about 730 hours of synthetic speech and about 100 hours of real speech. Its uniqueness lies in the parallel structure of the data, in which for each source recording there are synthetic versions with matching linguistic content. In addition, the researchers used the Montreal Forced Aligner (MFA) to produce TextGrid files documenting the temporal boundaries of the phonemes, thereby enabling a focused acoustic analysis of matching speech units.

For the empirical evaluation, a subset of 1,000 source recordings and the synthetic versions matching them was chosen. The researchers extracted spectral features of the LFCC and Log-Mel Spectrogram types, alongside Self-Supervised Learning representations based on wav2vec 2.0 and WavLM. By means of a symmetric Kullback–Leibler Divergence (KLD), the differences between the distributions of real and synthetic phonemes were measured, while Logistic Regression and Support Vector Machine classifiers were used to evaluate the discriminative ability between them. In addition, a Pearson correlation was computed in order to examine the relationship between the size of the statistical difference and the classification accuracy.

The study's results indicate significant variance between phonemes, generation systems, and representation types. In particular, diphthongs such as `OY` and `EY`, fricatives such as `SH` and `ZH`, and certain plosives showed features that distinguish between real and synthetic speech. In addition, differences were found between TTS and VC systems, with some of the VC systems showing relatively large spectral deviations from real speech. However, the relationship between KLD and Accuracy was not uniform: in traditional spectral representations, significant positive correlations were found in many cases, while in SSL representations weak and even negative correlations were also obtained. This finding emphasizes that a large statistical difference between distributions does not necessarily guarantee a high classification ability, and that the contribution of phonetic features depends on the representation and classifier chosen.

The study has several central limitations. The dataset is based on English speech from LibriSpeech and on only seven generation systems, and therefore the generalization ability of the findings to other languages, speakers, and recording conditions is not guaranteed. In addition, the phoneme boundaries were determined by means of automatic Forced Alignment, which may contain errors. The evaluation focuses mainly on KLD, correlations, and Accuracy at the phoneme level, and does not provide direct proof of an improvement in EER at the recording level or of generalization to synthesis systems not included in training. Likewise, the description of the experiments does not allow fully verifying the separation between speakers and source recordings in all the classification experiments.

In the context of the present project, PhonemeDF constitutes a significant candidate for expanding the training data by means of examples from different TTS and VC systems, thereby making it possible to examine the contribution of the diversity of generation methods to voice forgery detection. In addition, the phoneme-level information may serve as a basis for error analysis and for identifying acoustic features that the detection models struggle to cope with. However, using the dataset requires preserving the source associations between the real and synthetic recordings, checking overlaps, and separating speakers, in order to prevent information leakage and biases stemming from the data source. Its final suitability for training is also conditional on verifying the licensing terms and the available Metadata.

The central research conclusion is that synthetic-speech artifacts do not appear uniformly throughout the recording, and analysis at the phoneme level may provide information complementary to detection approaches based on global representations. However, the contribution of phonetic features and of training-data expansion to improving generalization requires a controlled empirical evaluation and cannot be inferred from KLD or Accuracy at the phoneme level alone.

Reference

Nallaguntla, V., Fursule, A., Kshirsagar, S., & Avila, A. R. (2026). *PhonemeDF: A Synthetic Speech Dataset for Audio Deepfake Detection and Naturalness Evaluation*. Proceedings of the Fifteenth Language Resources and Evaluation Conference (LREC 2026), 5905–5915.

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